Introduction to Agentic AI

Agenda

Why Agentic AI?

AI Agents vs Agentic AI

Evolution of Agentic AI

Reference Platform Architecture

Agent Patterns

Agent patterns contd.,

Why Agentic AI?

- LLMs evolving from passive tools to active agents Autonomous task orchestration
- Need for memory, reasoning, planning

AI Agents vs Agentic AI

An AI Agent is an individual intelligent system that:

Perceives → Decides → Acts to achieve a goal.

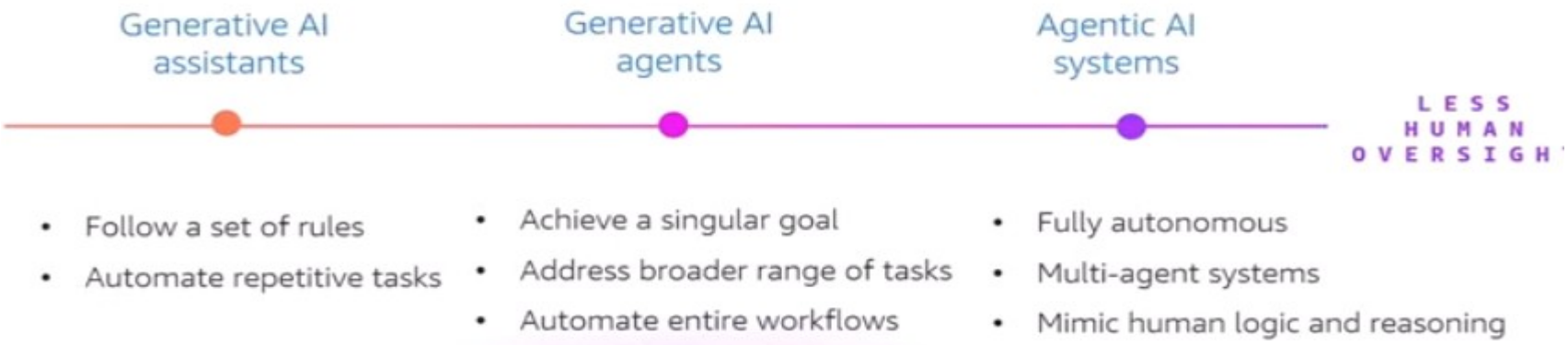
It usually works alone and focuses on completing specific tasks.

Agentic AI is a system-level paradigm where:

Multiple intelligent agents collaborate, plan, learn, and self-improve to achieve complex goals.

It is about building ecosystems of agents, not just one.

Evolution of Agentic AI



Reference Platform Architecture

Planner: Decomposes goals into actions or chains-of-thought. Can be rule-based or LLM-based (chain-of-thought prompts). Guides overall strategy

Memory: Often split into working memory (short-term conversation context) and persistent memory (long-term vector store)

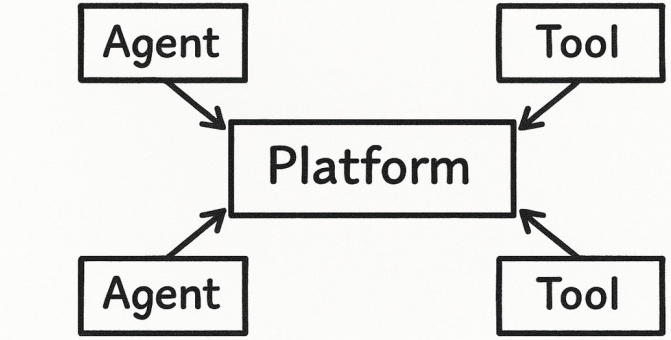
Execution/Tools: Agents execute plans by calling external tools/APIs (e.g., search, databases, web APIs)

Observability: Integrate logging, tracing, and metrics (e.g. via OpenTelemetry) to track agent decisions and performance

Environment: The “perception” layer – how agents receive input (user requests, sensor data, events) and output results (UI, API responses).

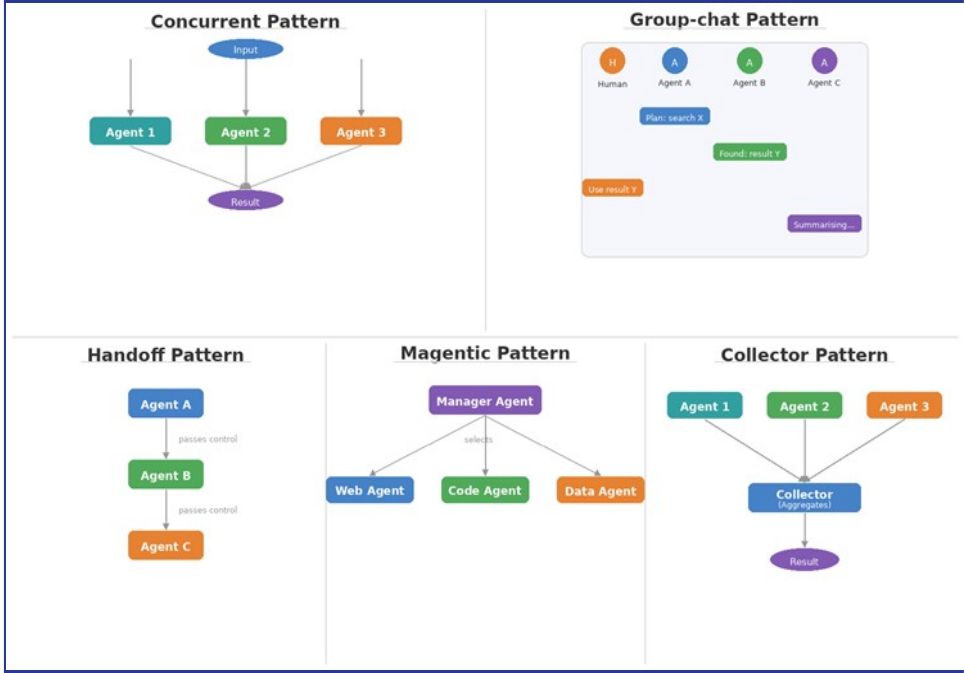
The Agent-Tool Ecosystem

As we move forward to the next stage of our discussion, let's take a
moment to reflect on the key findings that have led us to this point.
Our analysis has revealed some interesting trends and patterns that warrant further exploration.
Now, on this sixth slide, we'll delve into the implications
of these discoveries and how they might inform our decision-making process.



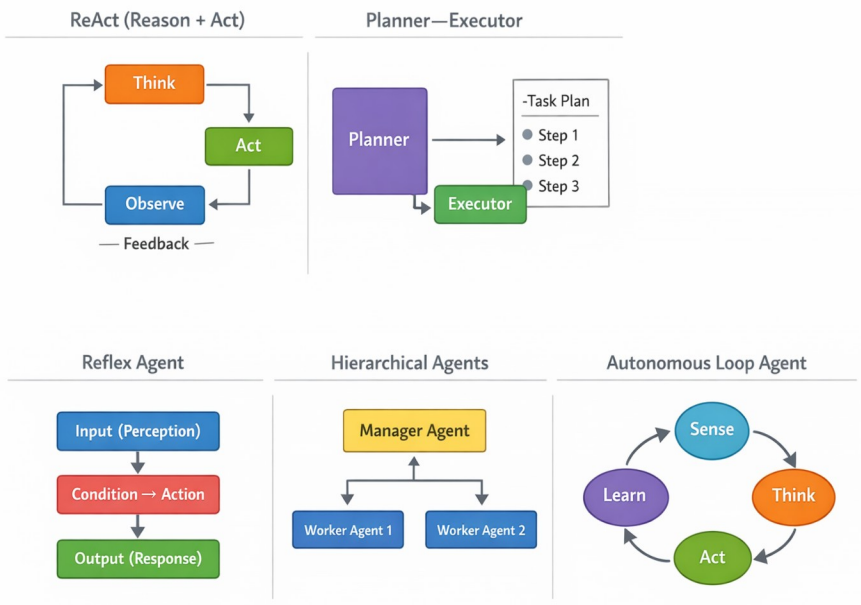
Agent Patterns

- Concurrent Pattern: Multiple agents work on the same input in parallel.
- Group-chat Pattern: Agents and optional humans interact via chat interface.
- Handoff Pattern: Agents dynamically pass the control
- Magentic Pattern: A central manager selects which special agent to invoke next
Collector Pattern



Agent patterns contd.,

- Planner–Executor: Planner creates a plan and executor performs steps. A type of Hand-off pattern.
- Reflex Agents: Respond to stimuli, they dont have memory. Used in chatbots, simple automation.
- Hierarchical Agents: High-level agent delegates tasks. This is also a type of magentic pattern.
- Autonomous Loop Agents:They have Continuous sense–think–act cycle



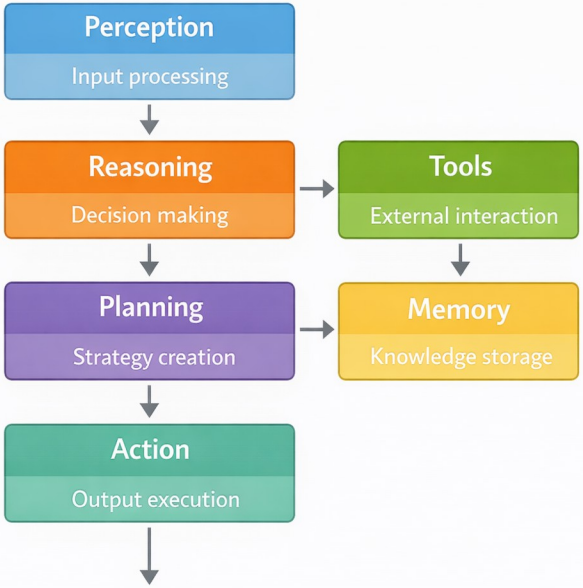
Agent Patterns

As we conclude our presentation, let's take a closer look at

Agent Patterns, which represent a crucial aspect of our overall architecture.

These patterns allow us to effectively manage and coordinate the

interactions between various components, ensuring a seamless and efficient experience.



Conclusion: Introduction to Agentic AI

- Agentic AI is a paradigm shift in artificial intelligence, focusing on autonomous agents that can achieve goals
- Understanding AI agents vs Agentic AI is crucial for effective implementation
- The evolution of Agentic AI has led to the development of reference platform architectures
- Agent patterns are essential for designing and building Agentic AI systems
- Real-world applications of Agentic AI have the potential to transform industries